SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1001	**Course Title:**	Software Engineering
CO Assessed:	CO1 -AT-2	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	27/10/2026	**Duration:**	60 minutes

Code of Conduct

I MOHAMED ZAMEER (1921260166) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: MOHAMED ZAMEER

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

AGILE SPRINT PLANNING AND USER STORY WORKSHOP

SMART WASTE PRODUCT DEVELOPMENT

1. Project Vision and Stakeholders

1.1 Project Vision

The vision of the Smart Waste Product Development project is to develop an intelligent waste management system that monitors waste levels in dustbins, improves waste collection efficiency, reduces overflowing bins, and promotes a cleaner environment using smart sensors and digital technology.

The system will use sensors to detect waste levels and provide notifications when bins need to be emptied. It aims to improve cleanliness, reduce manual monitoring, and support efficient waste collection.

1.2 Project Objectives

1. Develop a smart dustbin monitoring system.
2. Detect waste levels using sensors.
3. Notify waste collection staff when bins are nearly full.
4. Reduce overflowing dustbins and environmental pollution.
5. Improve waste collection efficiency.
6. Provide real-time information about bin status.
7. Develop a user-friendly monitoring interface.

1.3 Key Stakeholders

Stakeholder	Role and Responsibility
Project Owner	Defines project objectives and business requirements.
Product Manager	Maintains and prioritizes the product backlog.
Scrum Master	Facilitates Agile activities and removes obstacles.
Development Team	Designs and develops the smart waste management system.
Testing Team	Tests the system and verifies acceptance criteria.
Waste Collection Staff	Receive alerts and collect waste from bins.
Municipal Authorities	Monitor waste management operations.
Public Users	Benefit from cleaner surroundings and improved waste collection.
Maintenance Team	Maintains sensors, devices, and software.

2. Create Epics and User Stories

Epics represent large features that can be divided into smaller user stories.

Epic 1: Smart Waste Level Monitoring

User Story 1: Detect Waste Level

As a waste collection manager, I want the system to detect the waste level in a dustbin so that I can identify bins that require collection.

User Story 2: Display Bin Status

As a waste collection manager, I want to view the current status of each dustbin so that I can monitor waste accumulation.

Epic 2: Alert and Notification System

User Story 3: Receive Full-Bin Alerts

As a waste collection worker, I want to receive an alert when a dustbin reaches 80% capacity so that I can empty it before it overflows.

User Story 4: Identify Overflowing Bins

As a municipal supervisor, I want to identify bins that have reached their maximum capacity so that urgent collection can be arranged.

Epic 3: Waste Collection Management

User Story 5: View Collection Priority

As a collection supervisor, I want to view bins according to their waste levels so that I can prioritize collection activities.

User Story 6: Update Collection Status

As a waste collection worker, I want to mark a bin as emptied so that the system reflects its latest collection status.

Epic 4: Dashboard and Reporting

User Story 7: View Monitoring Dashboard

As a municipal administrator, I want to view all monitored bins on a dashboard so that I can understand the overall waste management situation.

User Story 8: Generate Waste Reports

As a municipal administrator, I want to generate waste collection reports so that I can evaluate collection performance.

3. Define Acceptance Criteria

Acceptance criteria specify the conditions that must be satisfied for a user story to be considered complete.

User Story 1: Detect Waste Level

- **Given** a smart dustbin contains waste,
- **When** the sensor measures the distance between the sensor and the waste,
- **Then** the system calculates and displays the estimated percentage of bin capacity occupied by waste.

Additional criteria:

- The displayed value must be between 0% and 100%.
- The system must indicate a sensor error when valid readings cannot be obtained.

User Story 2: Display Bin Status

- **Given** multiple dustbins are registered in the system,
- **When** the manager opens the monitoring dashboard,
- **Then** the system displays each bin's identifier, location, and latest available waste level.

Additional criteria:

- The dashboard must distinguish normal, nearly full, and full bins.
- Missing or outdated readings must be clearly identified.

User Story 3: Receive Full-Bin Alerts

- **Given** a dustbin is being monitored,
- **When** its measured waste level reaches or exceeds 80%,
- **Then** the system generates a notification for the assigned collection worker.

Additional criteria:

- The alert must identify the bin and its location.
- The system must record the alert status.
- Repeated alerts must be controlled to avoid unnecessary notifications.

User Story 4: Identify Overflowing Bins

- **Given** a monitored bin is nearly full,
- **When** its waste level reaches 100%,
- **Then** the system marks the bin as full and displays it as a high-priority collection requirement.

Additional criteria:

- The supervisor must be able to identify the affected bin.
- The full status must remain visible until a new valid reading indicates otherwise or collection is recorded.

User Story 5: View Collection Priority

- **Given** several bins have different waste levels,
- **When** the supervisor opens the collection list,
- **Then** the system displays the bins in descending order of collection priority.

Additional criteria:

- Full bins must appear before nearly full bins.
- Each entry must display the bin identifier, location, and waste level.

User Story 6: Update Collection Status

- **Given** a worker has emptied a dustbin,
- **When** the worker selects the bin and records the collection,
- **Then** the system updates its collection status and records the collection time.

Additional criteria:

- The record must identify the bin.
- The system must not automatically assume a bin is empty without a valid reading or an authorized collection update.

User Story 7: View Monitoring Dashboard

- **Given** the administrator has access to the system,
- **When** the administrator opens the dashboard,
- **Then** the system displays the total registered bins and the number of normal, nearly full, and full bins.

Additional criteria:

- The dashboard must use clearly labelled status indicators.
- The displayed counts must match the latest available bin records.

User Story 8: Generate Waste Reports

- **Given** the system contains waste monitoring and collection records,
- **When** the administrator selects a reporting period,
- **Then** the system generates a report containing the number of collections and the relevant bin status information for that period.

Additional criteria:

- The report must display the selected date range.
- Empty datasets must produce a clear message rather than an incorrect report.

4. Prioritize the Product Backlog

The product backlog contains all identified features and requirements. The backlog is prioritized according to customer value, technical dependencies, and implementation importance.

Priority levels:

- **P1 – High:** Essential for the first working version.
- **P2 – Medium:** Important improvements after core functionality.
- **P3 – Low:** Additional features that can be implemented later.

Priority	User Story	Business Value	Reason
P1	US1: Detect Waste Level	Very High	Core functionality of the product.
P1	US2: Display Bin Status	Very High	Enables users to monitor bin conditions.
P1	US3: Receive Full-Bin Alerts	High	Prevents overflowing bins and supports timely collection.
P1	US4: Identify Full Bins	High	Helps supervisors identify urgent collection needs.
P2	US5: View Collection Priority	High	Improves collection planning.
P2	US6: Update Collection Status	High	Maintains accurate collection records.
P2	US7: View Monitoring Dashboard	Medium	Improves overall visibility.
P3	US8: Generate Waste Reports	Medium	Supports long-term performance evaluation.

Prioritization approach: The MoSCoW method can also be applied:

- **Must Have:** Waste-level detection, bin-status display, and basic full-bin alerts.
- **Should Have:** Collection priority and collection-status updates.
- **Could Have:** Advanced dashboards and reporting.
- **Won't Have in the First Release:** Predictive waste generation analytics and route optimization.

The initial backlog focuses on delivering a working monitoring system before adding advanced management features.

5. Plan the Sprint Backlog

A sprint is a fixed development period during which the team delivers a usable increment of the product.

Sprint duration: 2 weeks

Sprint number: Sprint 1

Sprint focus: Develop the basic smart waste monitoring and notification functionality.

Selected User Stories

User Story	Description	Priority
US1	Detect waste level	High
US2	Display bin status	High
US3	Generate full-bin alerts	High

These stories are selected because they establish the minimum functionality needed to monitor bins and identify when collection is required.

Sprint Development Tasks

US1: Detect Waste Level

1. Identify suitable waste-level sensors.
2. Design the sensor and controller connection.

3. Develop sensor-reading software.
4. Calculate the estimated bin fill percentage.
5. Test readings at different waste levels.

US2: Display Bin Status

1. Design a simple monitoring interface.
2. Create a data structure for bin information.
3. Display bin identifiers and locations.
4. Display the latest waste-level readings.
5. Test status updates using sample readings.

US3: Generate Full-Bin Alerts

1. Define the alert threshold at 80%.
2. Develop the threshold-checking logic.
3. Create a notification mechanism.
4. Display the affected bin's identifier and location.
5. Test notifications at different fill levels.

Sprint Task Schedule

Period	Planned Activities
Days 1–2	Requirement review, sensor selection, and system design
Days 3–5	Sensor integration and waste-level detection
Days 6–8	Monitoring interface and bin-status display
Days 9–10	Alert development, integration, testing, and review

The schedule is provisional and may be adjusted based on team capacity, equipment availability, and technical challenges.

6. Estimate Effort Using Story Points

Story points estimate the relative effort, complexity, and uncertainty of a user story. The Fibonacci sequence is used because larger tasks involve greater uncertainty.

The estimation values used are 1, 2, 3, 5, 8, and 13.

Story-Point Estimation

User Story	Story Points	Justification
US1: Detect Waste Level	8	Requires sensor selection, hardware integration, calculation logic, and physical testing.
US2: Display Bin Status	5	Requires interface design, data handling, and display validation.
US3: Generate Full-Bin Alerts	5	Requires threshold logic, notification handling, and testing.
US4: Identify Full Bins	3	Mainly involves full-capacity status detection and prioritization logic.
US5: View Collection Priority	5	Requires sorting logic and a collection-list interface.
US6: Update Collection Status	3	Requires status updates and collection-time recording.
US7: View Monitoring Dashboard	5	Requires summary calculations and a dashboard interface.
US8: Generate Waste Reports	8	Requires data aggregation, date filtering, report formatting, and validation.

Estimation Technique: Planning Poker

Planning Poker can be used to estimate the user stories.

1. The team reviews each user story.
2. Team members discuss the complexity and technical requirements.
3. Each member selects a Fibonacci story-point value.
4. The estimates are revealed simultaneously.
5. Differences are discussed.
6. The team agrees on a final estimate.

For example, US1 receives 8 story points because it involves both hardware and software development, sensor calibration, and physical testing.

Total estimated backlog size: 42 story points.

Sprint 1 commitment: 18 story points, consisting of US1, US2, and US3.

The commitment is a planning estimate, not a guarantee. Actual sprint capacity depends on the team's experience, available time, equipment, and any technical difficulties.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

Sprint Goal: To develop a basic smart waste monitoring prototype that detects waste levels, displays the current status of dustbins, and generates alerts when the waste level reaches or exceeds 80% capacity.

This goal provides a clear and measurable target for Sprint 1.

7.2 Expected Deliverables

At the end of Sprint 1, the team is expected to deliver:

1. **Working Sensor Prototype:** A sensor-based setup capable of detecting waste levels.
2. **Waste-Level Calculation:** Software that converts sensor readings into an estimated fill percentage.
3. **Monitoring Interface:** A simple dashboard displaying bin identifiers, locations, and waste levels.
4. **Alert Functionality:** A notification mechanism that activates when the fill level reaches the defined threshold.
5. **Test Results:** Evidence that the sensor, dashboard, and alert functionality have been tested.
6. **Technical Documentation:** A brief description of the system architecture, components, and implementation.
7. **Demonstration:** A prototype demonstration showing how the system responds as the simulated or actual waste level increases.

7.3 Definition of Done

A user story is considered complete when:

- All associated development tasks are finished.
- The acceptance criteria are satisfied.
- The feature has been tested successfully.
- Identified critical defects have been resolved.
- The implementation is integrated with the rest of the prototype.
- Relevant documentation is updated.
- The completed functionality is demonstrated during the sprint review.

8. Sprint Review and Retrospective

8.1 Sprint Review

At the end of Sprint 1, the team demonstrates the prototype to the product owner and stakeholders.

The review evaluates:

- Whether the sensor detects waste levels correctly.
- Whether the dashboard displays the latest available readings.
- Whether an alert is generated at the defined threshold.
- Whether the delivered features meet their acceptance criteria.
- Which backlog items should be considered for the next sprint.

8.2 Sprint Retrospective

The development team discusses what went well, what problems occurred, and what improvements should be made.

Possible discussion points:

- Accuracy and reliability of sensor readings.
- Integration issues between hardware and software.
- Delays caused by equipment availability.
- Clarity of user stories and acceptance criteria.
- Improvements to testing and team coordination.

The team uses these findings to improve its process during the next sprint.

9. Future Sprint Plan

After Sprint 1, subsequent sprints can introduce additional features.

Sprint	Planned Features	Expected Outcome
Sprint 1	Waste-level detection, status display, and alerts	Basic monitoring prototype
Sprint 2	Collection priority and collection-status updates	Improved collection management
Sprint 3	Administrative dashboard and waste reports	Better operational visibility
Sprint 4	Performance improvements, expanded testing, and final integration	More reliable and complete prototype

The final sprint sequence should be adjusted according to the team's capacity, stakeholder feedback, and the results of earlier sprints.

10. Conclusion

Agile Sprint Planning helps organize the Smart Waste Product Development project into manageable development stages. By defining a clear project vision, identifying stakeholders, creating epics and user stories, specifying acceptance criteria, prioritizing the product backlog, and estimating effort with story points, the team can focus on delivering valuable functionality incrementally.

The first sprint concentrates on waste-level detection, bin-status monitoring, and full-bin alerts. These features establish the foundation for a smart waste management system that can reduce overflowing bins, improve collection efficiency, and support a cleaner environment.

The Agile approach also encourages regular testing, stakeholder feedback, teamwork, and continuous improvement throughout the development process.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25